

Vraj Patel

Systems Integration Engineer | Backend, Data, and Applied ML Infrastructure

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PROFILE

Systems integration engineer building dependable internal software, operational data pipelines, automation, and human-reviewed ML workflows. Strongest in Python/FastAPI, PostgreSQL, React/TypeScript, Docker, AWS, and recoverable long-running systems.

EXPERIENCE

Systems Integration Engineer — CU Boulder · Institute of Behavioral Science May 2025–Present

- Built a FastAPI, React/TypeScript, and PostgreSQL support platform over a current 10.8K-ticket Freshdesk corpus, with support-group IDs recorded for 96.7% of records.
- Developed and evaluated an unreleased BERT routing candidate on 6,137 training tickets, reaching 82.63% sealed-test accuracy and 0.7563 macro-F1.
- Reduced analytics visualization assets by 61% raw and 66% gzip by replacing Plotly with Recharts; automated a typical Active Directory cleanup from about 15 minutes to under one minute.

Research Appointment — Indian Parliamentary Data — Princeton University Jan 15, 2025–May 2026

- Built the scraping logic and orchestrated collection, speech and translation processing, PDF extraction, normalization, and structured outputs end to end under a separate paid Princeton appointment.
- Processed 40+ GB of campaign audio and thousands of parliamentary documents with resumable state and reviewable intermediate artifacts.

Lead Technical Research Assistant — CU Boulder · Institute of Behavioral Science Mar 2023–May 2025

- Built PyTorch and ResNet50 workflows for visual-bias analysis of New York Times COVID-19 imagery using OpenCV preprocessing and unsupervised embedding clusters.
- Built checkpointed Selenium, BeautifulSoup, and LegiScan pipelines for 15+ years of legislative data; caching and rate-limit handling cut redundant bulk-ingestion requests by approximately 40%.

Undergraduate Research Assistant — Satellite Telemetry — Purdue Data Mine Aug 2024–May 2025

- Through an L3Harris industry collaboration, trained Isolation Forest and LSTM models on synthetic NOS3 telemetry and co-presented satellite-cyberattack research to industry and government stakeholders.

Research Assistant, Data Engineering and ML — Studio Lab · CU Boulder Jan 21–Apr 13, 2024

- Designed WER-oriented comparisons across Deepgram, Google Speech-to-Text, Amazon Transcribe, and Azure Speech, plus translation comparisons across Google, AWS, and Azure services.

SELECTED CASE STUDIES

Operational Ticket Intelligence · *FastAPI, PostgreSQL, React, TypeScript, scikit-learn*

Core backend workflows, lifecycle-aware analytics, fail-closed readiness checks, and approval controls that keep predictions advisory and external updates bounded.

SeeMyRace · *Flask, Docker, PostgreSQL, DeepFace, PaddleOCR*

Contributed UI implementation, moderation-workflow design, ML-pipeline connectors, and bib-detection experiments to a private collaborative prototype.

EDUCATION AND SKILLS

University of Colorado Boulder · M.S. Computer Science + M.S. Engineering Management (in progress) · B.S. Computer Science, Business Minor (May 2025) · GPA 3.65/4.00 · Dean's List, 5 semesters

Skills: Python, TypeScript, JavaScript, SQL, PowerShell, Bash · FastAPI, PostgreSQL, React, Docker, AWS, Git, Linux · scikit-learn, PyTorch, Hugging Face, Pandas, OpenCV, Selenium